



TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: CP_SEP490

Duration time: From 01/09/2025 To 30/12/2025

Profession: Information Technology

Specialty: Software Engineering

Kinds of person make registers:

Lecturer: ☒ Students: ☐

1. Register information for supervisor

	Full name	Phone	E-Mail	Title
Supervisor 1	Trần Thị Tố Tâm	0905440132	tamttt14@fe.edu.vn	
Supervisor 2				

2. Register information for students

	Full name	Student code	Phone	E-mail	Role in Group
Student 1	Đỗ Thanh Tú	DE170525	0763794399	tudtde170525@fpt.edu.vn	Leader
Student 2	Kiều Văn Huy	DE170548	0763583081	huykvde170548@fpt.edu.vn	Member
Student 3	Mai Xuân Hiếu	DE170524	0896210393	hieumxde170524@fpt.edu.vn	Member
Student 4	Đinh Ngọc Tâm	DE170505	0935050703	tamdnde170505@fpt.edu.vn	Member
Student 5	Nguyễn Nho Quốc Đạt	DE170500	0762546820	datnnqde170500@fpt.edu.vn	Member

3. Register content of Capstone Project

3.1. Capstone Project name:

English: Online Medical Appointment Booking System

Vietnamese: Hệ thống đặt lịch hẹn khám bệnh trực tuyến

Abbreviation: **MABS**

3.2. Context (brief introduction)

Introduction:

The Online Medical Appointment Booking System (MABS) is a web-based platform designed to streamline the process of booking medical appointments. It connects patients with doctors and clinics, allowing users to search for available time slots, view doctor profiles, and schedule appointments online. The system aims to reduce waiting times, improve access to healthcare services, and enhance the overall patient experience.

Objectives:

The system enables patients to conveniently search for and book medical appointments online with doctors and clinics, based on specialty, availability, and location. It also supports video-based remote consultations, automated appointment reminders, a basic healthcare chatbot, online payment processing, and service feedback to improve overall quality.

Technology/algorithm:

- Front-end:
 - ReactTS
 - HTML, SASS
 - Bootstrap
- Back-end:
 - ASP .Net Core
 - SQL Server
 - Redis (caching)
 - Elasticsearch (for searching)
 - WebRTC (support video call)
 - SignalR (chat realtime)
 - SQS (push notifications)
- Cloud Platform:
 - S3 (storage static files)
 - CDN (caching)
 - EC2
- Version Control: Git, Github
- CI/CD Tools: GitHub Action

3.3. Summarize the contents to be researched and the expected outputs of the project:

1. **Patient:** Users search for medical services and book medical appointments
 - Registration & Login
 - Sign up using email, phone number, Google/Facebook
 - OTP phone verification
 - Email verification
 - Logout
 - Search & Book Appointments
 - Enter date, location for search and select desired specialty or doctor
 - Choose hospital/clinic based on location and availability
 - Send requests to making an appointment
 - Receive confirmation notifications from hospital/clinic
 - Cancel/reschedule appointment
 - Get calendar reminders
 - Filter & Select Doctors
 - Search by ratings, consultation fees, specialization, or location
 - View doctor profiles, years of experience, and patient reviews
 - Remote consultation via Chat & Video Call with Doctors
 - Integrated real-time chat for consultation or appointment preparation
 - Video call for remote consultation (Telehealth)
 - Online Payment & Invoice Management

- Support for MoMo, VNPay (make a deposit according to the service fee or complete the full payment)
- View and download payment history and electronic invoices
- Review & Feedback
 - Rate doctors and services (1–5 stars)
 - Write reviews
 - Report unprofessional conduct or poor service
- Explore Services, FAQs & Health Content
 - View health blogs, articles, and handbooks
 - View available medical services (general check-up packages, blood tests)
 - View FAQs
- Smart Suggestions & Assistance
 - Receive doctor or department suggestions based on symptoms, location, or medical history
 - Use chatbot for basic health advice, appointment guidance, and FAQ support
 - AI-based symptom recognition from text input (e.g., “fever, headache” → suggest Internal Medicine)
- Manage personal information
 - Allow patients to modify their personal details (e.g. phone number, email, address, password)
 - View medical history
 - View payment history

2. **Doctor:** Manage examination schedules and consult with patients both in-person and online.

- Authentication
 - Sign in using your email or phone number
 - Sign out
- Manage Personal Profile
 - Allow doctor to modify their personal details (e.g. phone number, email, address, password)
- View, Cancel Appointments
 - View upcoming medical appointments
 - Cancel medical appointments (approved by the staff of the hospital/clinic)
- View Appointments History
 - Access patient records from the appointment list or manual search
 - Filter by date, price and name of patient
- Reply To Patient Reviews
 - Respond to patient feedback
- Online Patient Consultation (Video Call/Chat)
 - Remote medical examination via camera
 - Chat with the patient
- Auto-generate Medical Notes
 - Convert examination results from the doctor's voice into text automatically entered into the medical examination results

3. **Admin:** Manages user accounts, healthcare data, services, schedules, and system content. Generates reports and oversees overall system operations.

- Login
- User account management (doctors, patients, hospitals/clinics)
 - Add, update, or remove doctor, patient, and medical hospital/clinic accounts

- Activate/deactivate user accounts
 - View full user profiles
- Specialty category management
 - Create, edit, and delete medical specialties (e.g., Cardiology, Pediatrics)
 - List specialties for easier filtering
- Medical hospital/clinic management
 - Manage hospital/clinic information (name, address, contact, status,...)
 - Add new hospital/clinic or update existing ones
- Statistics and report generation by time (day, month, year, by hospital/clinic, by specialty, etc.)
 - View dashboards with charts for appointments, revenue, and feedback
 - Generate reports by day, month, year
 - Filter reports by hospital/clinic, doctor, specialty, or service
 - Export reports to Excel or PDF
- Service management
 - Create, update, delete and manage healthcare services
 - Approve hospital/clinic service registration request
- Appointment schedule management
 - View and manage all patient appointments across doctors and hospitals/clinics
 - View appointment analytics (total, by hospital/clinic, by doctor, etc.)
- Blog content management (news, blog)
 - Create and manage health-related content (blogs, news articles)
 - Categorize content by topic or type (handbook, health life, ...)
 - Moderate posts before publishing
- FAQ management
 - Create, edit, and categorize frequently asked questions
- Banner management
 - Upload and manage homepage and internal banners
 - Set banners to promote services, news, or announcements
 - Set banner visibility by page, time range

4. **Staff (Hospital/Clinic):** Manage the doctors of that hospital/clinic and be responsible for the doctors' schedules

- Authentication
 - Sign in using email or phone number
 - Sign out
- Hospital/Clinic information management
 - Update hospital/clinic information (name, address, phone number, email, website, introduction, images, operating license).
- Doctor Management
 - Add, delete, or update doctor information (profile, specialties, work schedule).
 - Assign doctors by specialty or medical hospital/clinic.
- Specialized management
 - Add, update or delete hospital/clinic specialty categories.
- Health service management
 - Add, update or delete services
 - Request service prices when assignment
 - Update service prices and detailed descriptions.
- Manage medical appointment schedule
 - View and adjust hospital/clinic doctor appointments.
 - Approve or cancel appointments (in special cases, e.g. equipment maintenance, holidays).
- Statistics and reports

- View hospital/clinic performance reports (number of patients, revenue, doctor, or time).
 - Export reports to Excel, PDF for analysis or storage.
 - Promotion Management
 - Create and manage discount and promotion programs for medical services.
 - Promotional program quality assessment
 - Blog content management (news, blog)
 - Creating, updating, or deleting blogs content (request admin approval).
5. **Guest:** Can view hospital/clinic info, doctors, services, blogs, FAQs, and promotions. Cannot book or pay without registration.
- View Hospital//Clinics Information
 - Browse list of hospitals/clinics with basic details (name, address, phone, email, website, description, images)
 - Search for Doctors or Specialties
 - Search doctors by name, specialty, or hospital/clinic
 - View basic doctor profiles (name, specialty, affiliated hospital/clinic)
 - Cannot access detailed schedules or patient reviews
 - View Medical Services
 - Browse available services (e.g., general checkup, blood test, telemedicine)
 - Read service descriptions
 - Cannot book or pay for services
 - Access Blog & News
 - Read blog posts, health guides, and news articles posted by hospitals/clinics or administrators
 - View announcements and promotional banners
 - Read FAQs
 - Access frequently asked questions about the system and booking process
 - Use Health Chatbot Assistant
 - Get guidance on how to use the system
 - Ask basic health questions and receive preliminary advice
 - Get suggestions for relevant specialties based on symptoms
 - View Promotions
 - See ongoing discounts and promotional offers
 - Sign Up for an Account
 - Access guidance for registering a patient account to unlock full system features
 - Request Hospital/Clinic Partnership
 - Fill out a form to propose adding new hospital/clinic partners to the platform

Guarantee compatibility across multiple platforms and devices:

● Website application:	☒		● Mobile application:	☐	Other	☐
● Game:	☐		● Research:	☐		

3.4. Plan for applying AI to the project

N Plan to use AI to optimize work efficiency

Including choosing the right AI tools to support different stages in the software development process such as: requirements gathering, design, programming and testing, deployment, project management....

N Plan to integrate AI services to enrich system functionalities

Including integration of external AI services (OpenAI API, Google Gemini API, Azure AI Services, etc.) or intermediary platforms (Google AutoML LlamaIndex,...) to enhance the system's capabilities.

☐ *Plan to develop in-house AI modules*

Includes building and refining AI models to suit the project's unique needs, focusing on activities such as data processing, model training, evaluating results, and continuous improvement.

3.4. Expected features

1. Functional features:

- **Appointment Booking:**
The patient searches for available doctors or hospitals/clinics by specialty, location, and date → Selects an available slot → The system sends booking requests to the selected hospital/clinic → The hospital/clinic staff reviews the appointment → Confirms or declines → The patient is notified via email/SMS/in-app.
- **Online Consultation (Telehealth):**
The patient and doctor join a video call via the system → The doctor reviews the patient's symptoms → The doctor provides consultation and updates medical notes during the session.
- **User Authentication & Verification:**
Sign up/login via email, phone, or social accounts → Phone OTP and email verification required.
- **Payment & Review:**
The patient pays for services through supported payment gateways (MoMo, VNPAY) → The patient leaves a review and rating for the doctor and service quality.
- **Doctor Profile Management:**
Doctors update their profiles → View their schedule → Access patient history → Respond to patient reviews.
- **Hospital/Clinics Details Management:**
Hospital/Clinic staff manage doctor assignments, services, and schedules → Approve or cancel appointments → Update hospital/clinic information.
- **Blog & Health Information Access:**
Admin and staff post health tips, news → Patients and guests access blog content and announcements.
- **Smart Search & Recommendations:**
The patient enters symptoms into the system → AI suggests appropriate specialties or doctors → The patient is guided to book a suitable service.
- **Guest Browsing Mode:**
Guests can view hospital/clinic information, doctor lists, available services, and health blogs → Encouraged to register for full system access.

- Reports & Promotions:
Admin and staff view performance metrics (appointments, revenue, patient feedback) →
Export reports → Manage promotional programs and track their effectiveness.

2. Non-functional features:

- Performance:
The system should ensure fast response times, especially during peak usage (e.g., under 2 seconds for main operations).
- Scalability:
The system should support a growing number of users, appointments, and hospitals/clinics without performance degradation.
- Availability:
The platform should maintain high uptime (99.9%) to ensure uninterrupted access for patients and doctors.
- Security:
Ensure data privacy and protection of patient records through encryption, authentication, and role-based access control.
- Usability:
The system should have a user-friendly interface that is intuitive for both medical staff and patients.
- Maintainability:
The codebase should be modular and well-documented to hospital/clinic easy updates and maintenance.
- Portability:
The system should be accessible across different devices and browsers (desktop, tablet, mobile).
- Reliability:
The system should function correctly under expected conditions and handle errors gracefully.
- Localization:
Support multiple languages (e.g., Vietnamese and English) to reach a wider user base.

Supervisor (If have)
(Sign and full name)

Da Nang, / /2025
On behalf of Registers
(Sign and full name)